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Conceptual core design study for Indonesian Space Reactor (ISR)

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ABSTRACT

Space exploration is very important for the future of the earth and human beings as it may eliminate earth overpopulation and overcome diminishing of earth resources. One of the obstacles of the space exploration mission is the energy source for the spacecraft. One alternative is using a nuclear reactor as an energy source in spacecraft. A conceptual design of Indonesian Space Reactor (ISR) has been carried out to explore such a possibility. ISR is a liquid metal Na-78 K cooled space reactor with a fast neutron spectrum. It is designed to provide at least 500 kW_{th} power for operating time more than 10 years at full power. The reactor uses 55% high-enriched uranium nitrate as fuel. The ISR hexagonal core is comprised of 61 fuel pins and is designed in the form of a hollow cylinder with an individual cooling channel in each fuel pin. The reactor is also equipped with spectral shift absorbers (SSA) made of Re and Mo-30Re alloy to control the reactivity. Neutronic calculations have been performed to obtain optimum design parameters without compromising safety requirements. These design parameters include variation in uranium enrichment, reactor dimension, reflector thickness and control drum (absorber) design and dimension. The accepted reactor design has an excess reactivity of 4023 ± 9 pcm and shutdown margin of 4852 ± 9 pcm and the reactor is estimated to have a lifetime of 28 years. The temperature and void reactivity coefficients are all negative, implying inherent safety. Several accident scenarios were also considered in this work, both during launch failure and normal operation. It is found that to keep the reactor subcritical for a submerged reactor following a launch failure, the reflector segment should be discarded. Meanwhile, some portions of fuel pins should be removed from the core during operational accidents.

1. Introduction

The development of space science is essential for the future. NASA plans to return 5 to 10 astronauts to the moon in 2020 (Hatton and El-Genk, 2009). Space exploration is not limited to experimental missions, where mineral mining is an interesting issue to do. This activity certainly requires large electrical or thermal energy. One of the obstacles of the space exploration mission is the energy source for spacecraft needs.

Space reactors have unique characteristics such as high unit mass power, low cost, and strong environment adaptability; so that the space reactor is crucial for aerospace industry (Yuan et al., 2016). The use of space reactors as a source of energy in space can be justified when there is no choice of other energy sources (El-Genk, 2009). Such usage is inline with the United Nation principle 3 to reduce the amount of radioactive material in space, the use of nuclear energy in space is only permitted if there are no other alternative energy sources for space missions for acceptable reasons" (United Nations General Assembly, 1992).

In the mission of deep space exploration that keeps away from the sun, the intensity of sunlight decreases, so that solar panels cannot be used as energy generators. This condition makes no alternative energy generation other than space reactors. In general, there are no big differences between a space reactor and a typical reactor. It only needs some design adjustments such as mass, dimensions, power, and the operating time of the reactor. By considering the long space reactor operation, compactness, low mass and independent on sunlight, the space reactor becomes very promising to be used as a power plant in space missions, especially for spacecraft on the deep space exploration mission.

BUK, TOPAZ, and SNAP-10 were space reactors that used Highly Enriched Uranium (HEU) as fuel, and used 90%–96% enriched U-235 (El-Genk, 2009). In its development, some space reactors such as HOMER, SAIR, S4, and KRUSTY also used HEU as fuel (El-Genk and Tournier, 2004; King and El-Genk, 2006; Mencarini and King, 2018; Poston, 2000). The use of HEU as fuel aims to eliminate the use of moderators so that the reactor is compact and lightweight. The absence of the moderator made the reactor have a fast spectrum of neutrons.

It is not an absolute requirement that space reactors cannot use moderators, like SNAP-10 which used uranium-zirconium hydride

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(UZrH) and yet the reactor was compact (Stephenson and Blancquaert, 2008). Meanwhile, the presence of the moderator will limit the reactor operating temperature as ZrH moderator limits the reactor operation < 900 K which eventually will reduce the thermal efficiency. Such limitation makes that the thermal neutron reactor more suitable for low power (< 150 kW_{th}) and short missions (< 2 years), while fast neutron reactors are more suitable for high power and long missions (El-Genk, 2009).

The use of Low Enriched Uranium (LEU) as fuel is still possible, but the addition of the total reactor mass and the dimensions of the reactor must be taken into consideration. The use of LEU on KRUSTY produced a total reactor mass 2.1 times greater than the total mass of HEU-KRUSTY (Mencarini and King, 2018). The study of the use of LEU in the space reactor that has been carried out produced a reactor design with a total reactor mass of 240 kg and operating time of 15 years, but its thermal power is only 5 kW_{th} (Lee et al., 2015b). As a comparison, the SAIR design with HEU fuel has a total reactor mass of 424 kg, operating > 7 years, and has a thermal power of 407 kW_{th} (El-Genk and Tournier, 2004). The design of S4 with HEU fuel has a total reactor mass of ~600 kg, operating time > 7 years, and has a thermal power of 471 kW_{th} (King and El-Genk, 2006). Other research shows that for the same power and length of reactor operation, the use of LEU makes the reactor heavier and larger in dimension; compared to the use of HEU (Lee et al., 2015a). These reasons make space reactor with HEU fuel more promising than using LEU.

In the United Nation principle 3 parts of the nuclear reactor, space reactors must use HEU as fuel (United Nations General Assembly, 1992). On the other hand, in the treaty "the Non-Proliferation Treaty (NPT)" which has been signed by 191 countries suggests that the ban on the use of HEU in civilian applications (International Atomic Energy Agency, 2017, 2005). In this provision, HEU is defined as uranium with enrichment ${}^{233}_{92}U + {}^{235}_{92}U \geq 20\%$ (International Atomic Energy Agency, 2001). This issue should not be a barrier for the space reactor to use HEU as fuel, given that the spirit of the NPT is the disarmament of nuclear weapons and nuclear for peace (International Atomic Energy Agency, 1970).

Spectral Shift Absorbers (SSAs) have relatively small neutron absorption cross-section in the fast spectrum and large absorption cross-section in the thermal regions. Fig. 1 shows a neutron absorption cross-section accessed from the IAEA's Nuclear Data Center (International Atomic Energy Agency, 2018). In the scenario of a launch accident, when the fast reactor is submerged in seawater or wet sand, it will cause a moderated neutron and an increase in reactivity. The use of SSAs is beneficial in conditions of this launch accident, when neutrons are moderated (Webb and Charit, 2015). It is seen in Fig. 1, that gadolinium (Gd) is better as SSAs, but Gd has a lower melting point, so it can limit the reactor operating temperature. In the designs of SAIR, S4, and ScoRe, rhenium is used as SSAs in the form of Mo-%Re alloy (El-Genk and Tournier, 2004; Hatton and El-Genk, 2009; King and El-Genk, 2006).

The selection of coolants in the space reactor has several considerations such as temperature limits, coolant phase, and auxiliary system addition. The use of gas as coolant does facilitate the operation of the reactor. The negative side of the gas coolant is the high operating pressure so that it requires a thick reactor pipe and vessel which increases the total mass of the reactor. In addition, the gas coolant is prone to leak, so it is feared that the coolant mass flow rate in the reactor will decrease. To overcome this problem, an additional system is needed in the form of a gas injection system taken from gas storage. Adding a system like this will definitely make the reactor more complicated and increase the reactor mass.

The use of sodium as a coolant significantly helps reduce the spread of radioactive substances during an accident as sodium will return to the solid phase when it is cold. The obstacle of using sodium in a space reactor is the addition of an auxiliary system to convert sodium from a solid phase into a working fluid. Besides that, sodium heat pipe is a very promising candidate for passive cooling systems, for example used in LEGO, HOMER-15, SAIR, and KRUSTY reactor designs (Bess, 2008; El-Genk and Tournier, 2004; Mencarini and King, 2018; Poston, 2000). Several non-nuclear testing have been carried out at KRUSTY and show the validation of the operating system and controls (including the sodium heat pipe system) (Briggs et al., 2018). Even so, no single space reactor that uses sodium heat pipes ever flew.

The use of Na-78 K proved successful in the BUK, TOPAZ-I and SNAP-10 reactor (Bennett, 1989; El-Genk, 2009). Na-78 K has a liquid phase at room temperature and this makes Na-78 K coolant easy to operate. However, the Na-78 K boiling point is lower than sodium boiling point. The ease of the start-up process and proven success are what underlies the use of Na-78 K as an ISR coolant.

In this article, an Indonesian Space Reactor (ISR) reactor design will

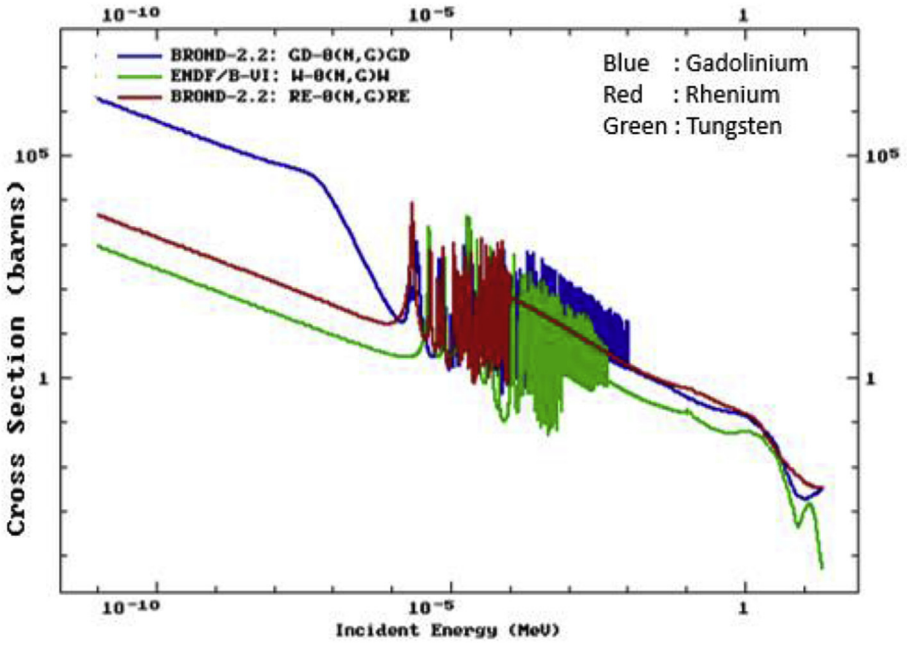


Fig. 1. Neutron absorption cross-sections for gadolinium, rhenium, and tungsten.

be described. In this work, to overcome the issue between NPT agreement and the United Nations about space reactor requirements, this new space reactor is designed using HEU but with a lower level of U-235 enrichment than the existing designs. Uranium Nitrate (UN) with natural nitrogen is used as fuel to reduce the cost of making reactors, although larger absorption cross-section of nitrogen-14 can lead to an increase in the mass and dimensions of the reactor. These two points are aimed to accelerate the commercialization of space reactor. Moreover, ISR is designed with a fuel pin geometry in the form of a hollow cylinder or an annulus, and coolant flows inside the annular channel of each fuel pin. Such fuel pin design makes it possible for the cooling flow rate to vary by each fuel pin, so it is expected that the core temperature distribution in the radial direction can be evenly distributed. In addition, neutron parameters will be presented. Furthermore, some criticality calculations in reactor accident conditions will be displayed, and a new idea about the disposal of damaged fuel pins from the reactor system during operations is also discussed. The ISR design will be explained in the next section.

2. Methodology

2.1. Indonesian Space Reactor (ISR) overview

Indonesian Space Reactor (ISR) is a liquid metal Na-78 K cooled space reactor with a fast neutron spectrum. It is designed to provide at least 500 kW_{th} power for operating time more than 10 years at full power. ISR uses UN fuel with U-235 enrichment of 55%. This enrichment level is relatively lower than the other existing space reactor designs, makes ISR safer from exposure to NPT issues. This lower enrichment also makes the production cost of the reactor cheaper than the higher U-235 enrichment. The initial design of the ISR has reactor inlet and outlet temperatures of 750 K and 950 K. The Na-78 K boiling point at ~1000 K makes the maximum temperature of the reactor operation is limited to that value, to avoid the formation of bubbles in the coolant. ISR's thermal energy conversion systems to electric can use static or dynamic thermal to electric conversions. The material data at 25 °C used in ISR is presented in Table 1.

Fig. 2 shows the ISR fuel pin geometry which is designed to have cooling channels in each fuel pin. Such a design allows the value of the Na-78 K coolant mass flow rate at each fuel pin to vary and to be adjusted to the heat generated by each fuel pin. Such a scheme is expected to make the distribution of the core temperature in the radial direction tends to be evenly distributed. ISR coolant pipe is made of rhenium with diameter inlet and a thickness of 0.6 cm and 0.08 cm. In this initial design, the cooling channel on each fuel pin has the same diameter inlet. It is possible during the development of the next design, different inlet diameter of each fuel might be used.

Mo-30Re cladding is hexagonal with a height of 43.2 cm. The hole in the shape of a cylindrical in cladding with a diameter of 3.265 cm contains rhenium discs, helium gas, fuel, and axial reflectors (see Fig. 2). Rhenium is used as SSAs in the form of Mo-30Re alloy and rhenium discs. Rhenium will help to keep the reactor subcritical when it is submerged in freshwater or seawater during accidents. The high melting point of rhenium (~3180 °C for rhenium and ~2790 °C for Mo-30Re alloy) is very suitable for high operating temperature reactor. On the other hand, rhenium is heavier than stainless steel that it could increase the reactor mass. Rhenium disc, fuel, and axial reflector are hollow cylinders. The fuel pin installation process can be done by welding the inner axial reflector (7 cm) at the end of the cladding. Furthermore, the cooling channel made of rhenium is installed and followed by rhenium disc, fuel, rhenium disc, and compression spring. Helium gas is injected at 1 atm pressure to fill the gap in the fuel pin. Finally, a 6-cm outer axial reflector is installed and welded. In the compression spring, it will form a helium gap as high as 5 cm.

Axial reflectors are designed without a gap with the cladding so that it is expected to hold helium gas, so it does not leak out of the fuel pin. The 5 cm helium gap where the compression spring is located is used as a place for the fission product gases. UN fuel and rhenium disks are designed to have a gap with the cladding inner surface of 25×10^{-4} cm. It is expected that the fission product gases will go through this gap to the helium gap.

Fig. 3 shows a simple scheme of the reactor installation process. As many as 61 fuel pins are arranged into hexagonal Mo-30Re core block as thick as 0.1 cm. The blank section between the arrangement of fuel pins and the core block is filled with Mo-30Re alloy. Multifoil insulation of 0.25 cm thick is installed on the outer surface of the core block to reduce heat transfer from the core to the radial reflector. The high temperature on the radial reflector is feared to interfere with the control drums performance.

There are two choices in the installation of radial reflector + control drum (see Fig. 3): (a) Radial reflector in the form of a whole hollow cylinder is attached to the core and has 6 holes where the control drums will be installed. (b) The radial reflector is divided into 6 segments that have been installed with a control drum. Finally, the reactor is fitted with an MA-965 steel drum with a thickness of 0.1 cm as reinforcement. Fig. 4 shows the ISR radial cross-section, and Table 2 shows the ISR mass estimate.

Each fuel pin has an individual cooling channel. These channels are arranged in such a way that the fuel pin can be removed from the reactor system when it breaks during operation. A new idea about the disposal of damaged fuel pin from the reactor system during operations is proposed in this work. A scheme like this makes the reactor still able to operate when there is damage to one of the fuel pins. In the next section, studies on neutronic aspects will be discussed. Furthermore, the scheme of how to remove the fuel pin from the reactor system and the issue of adding waste in space due to fuel pin disposal will be discussed in other work.

Table 1
Material properties at 25 °C.

Parameter	Material	Theoretical Density (g/cc)
Coolant	Na-78 K	0.732
Coolant pipe	Rhenium (Re)	21.020
Fuel	UN (55% U-235)	14.264
Cladding	Mo-30Re	11.963
Core clock	Mo-30Re	11.963
Multifoil insulation	Alumina paper type APA-1	0.200
Reflector	BeO	2.855
Absorber	B_4C (80% B-10)	2.510
Reinforcement material	MA-956 steel	7.250

2.2. Calculational scheme

The calculations in this study were carried out using the software SCALE version 6.1 with the 238-groups ENDF/B-VII.0 library (Oak Ridge National Laboratory, 2011). Criticality calculations were performed using the Criticality Safety Analysis Sequence with KENO-VI (CSAS6) control module, and the CENTRM module was used as cross-section processing. KENO-VI uses the Monte Carlo Method. The criticality simulation used 100,000–140,000 particles with 550 generations and 50 generations were skipped to reach standard deviation on every simulation $\leq \pm 10$ pcm.

The TRITON control module was used to perform burn-up calculations and the CENTRM module was again used as cross-section processing. TRITON also included Monte Carlo depletion capabilities using KENO-VI that has powerful 3D geometric representations for depletion calculation (Oak Ridge National Laboratory, 2011). For burn-up calculations, 120,000 particles were used with 330 generations in which 30 generations were skipped. All material depletion (Rhenium, UN, Mo-30Re, Re, BeO, B_4C, etc) is calculated in this scheme.

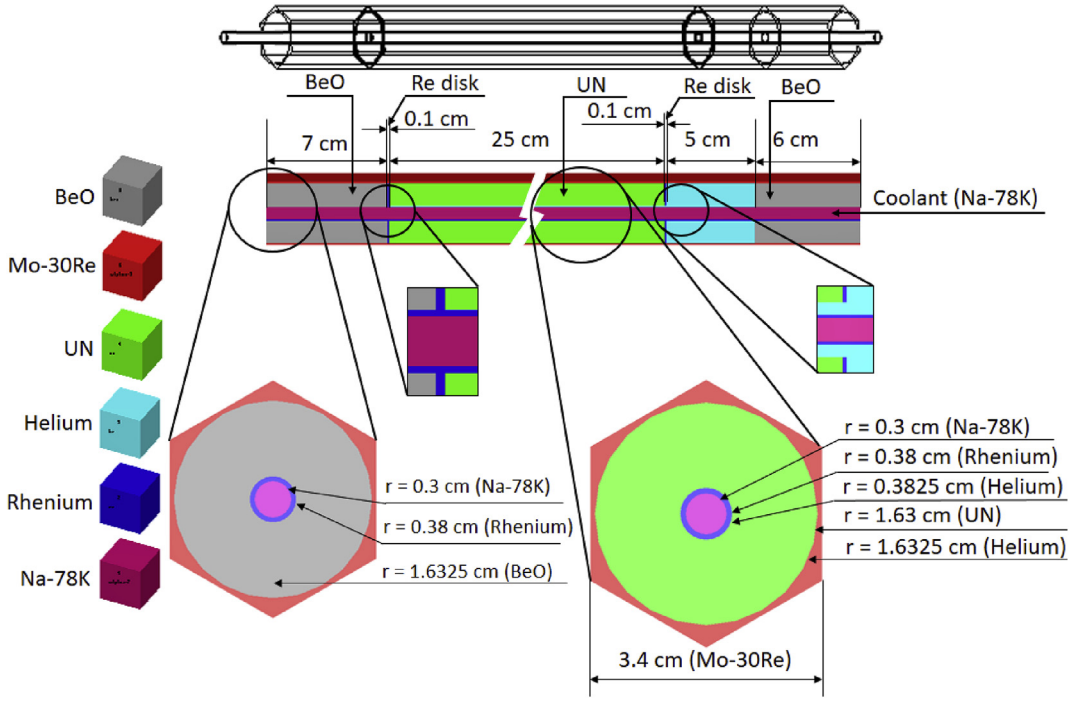


Fig. 2. Geometry and material used for the ISR fuel pin.

2.3. Materials input

The materials data used in this study are presented in Table 1. The calculation of UN theoretical density at temperature $298 \leq T \leq 2523$ K follows Eq. (1). Na-78 K theoretical density is calculated using Eq. (2), with a theoretical density of Na and K calculated using Eqs. (3) and (4), respectively (Institute for Heat and Mass Transfer in Nuclear Power, 2006).

$$\rho_{UN}\left(\frac{kg}{m^3}\right) = 14420 - 0.2779T - 4.897 \times 10^{-5}T^2 \tag{1}$$

$$\frac{1}{\rho_{Na-78K}}\left(\frac{m^3}{kg}\right) = \frac{0.22}{\rho_{Na}} + \frac{0.78}{\rho_K} \tag{2}$$

$$\rho_{Na}\left(\frac{g}{cm^3}\right) = 0.89660679 + 0.5161343(T \times 10^{-3}) - 1.8297218 (T \times 10^{-3})^2 + 2.20162470(T \times 10^{-3})^3 - 1.397563400 (T \times 10^{-3})^4 + 0.44866894(T \times 10^{-3})^5 - 0.057963628 (T \times 10^{-3})^6 \tag{3}$$

$$\rho_K\left(\frac{g}{cm^3}\right) = 0.90281376 - 0.16990711(T \times 10^{-3}) - 0.26864769(T \times 10^{-3})^2 + 0.50568188(T \times 10^{-3})^3 - 0.465379120(T \times 10^{-3})^4 + 0.20378107(T \times 10^{-3})^5 - 0.034771308(T \times 10^{-3})^6 \tag{4}$$

2.4. Reactor design

There are several main targets in the ISR design: (1) The reactor has

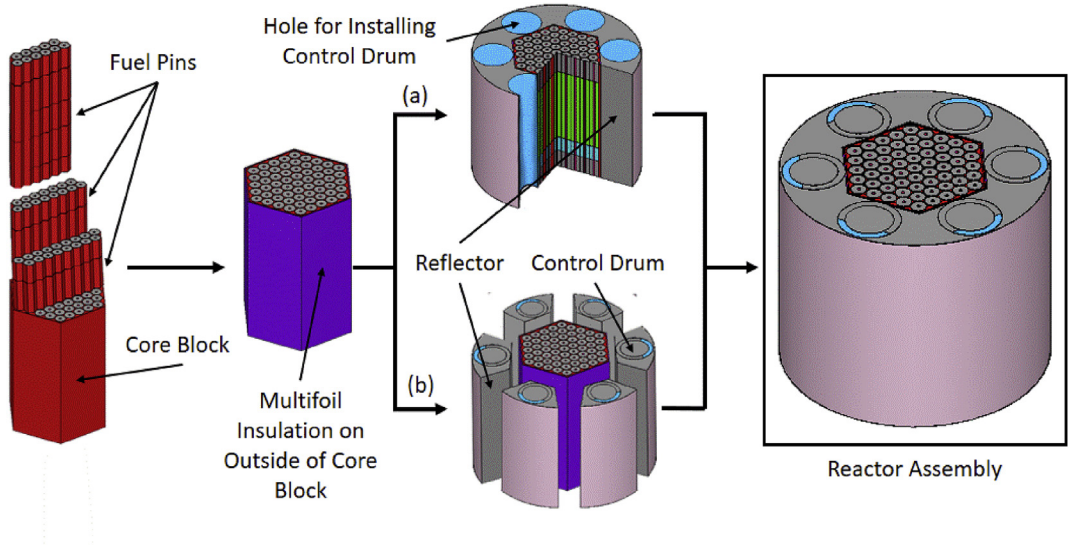


Fig. 3. Arrangement of the ISR assembly.

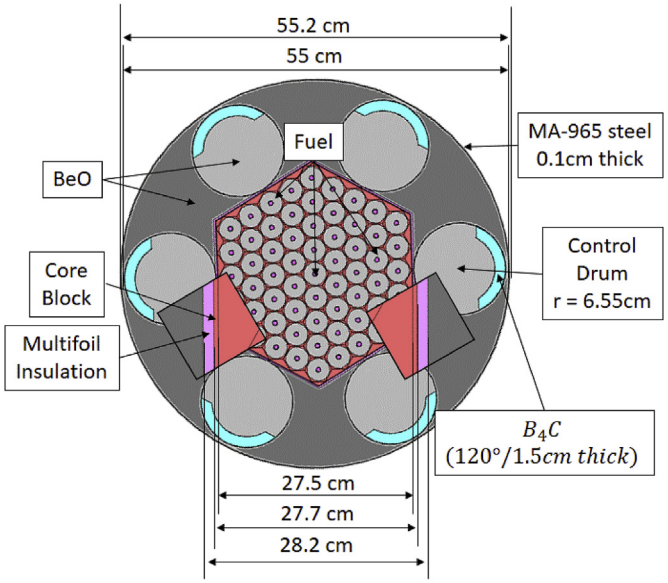


Fig. 4. Radial cross-section of the ISR.

Table 2
Estimation of the ISR mass.

Parameters		Mass (kg)
Fuel pin	coolant pipe (Re)	0.23
	fuel (UN)	2.55
	axial reflectors (BeO)	0.26
	Re caps	0.03
	Na-78 K	0.01
	cladding (Mo-30Re)	0.13
Total fuel		155.65
Core (61 fuel pins)		195.74
Core block		50.80
Multifoil Insulation		1.25
Radial reflector + 6 control drums		194.37
Reinforcement material		4.96
Reactor total mass		**447.12**

ample reactivity for full power operation for ≥ 10 years with thermal power of at least 500 kW_{th}. The excess reactivity is expected to overcome negative reactivity feedbacks from temperature, fuel burn-up, and fission products. (2) In normal and cold clean conditions at 25 °C, the reactor has a shutdown margin ≥4000 pcm to maintain the reactor subcritical during launch. (3) The reactor remains in subcritical conditions in the worst accident scenarios with a shutdown margin ≥2000 pcm. This condition is described when the condition of the reactor is cold clean, shutdown (i.e., all control drums are functioning), and submerged in freshwater or seawater. All the gaps in the cooling channels and the helium gap in the reactor is assumed to be filled with water and the reactor is surrounded by about 2 m of water in all directions, thus increasing neutron moderation. (4) The reactor has an independent cooling system for each fuel pin to achieve even distribution of radial heat. (5) Enrichment of U-235 is less than 60%. The design workflow is shown in Fig. 5.

The Na-78 K temperature used is the average value of the reactor inlet and outlet temperature, which is equal to 850 K. In this initial study, the calculation of the temperature distribution of one fuel pin was carried out by an analytical method, assuming the reactor power was evenly distributed. From this calculation, it was obtained that the average UN temperature was 950 K. More detailed thermal hydraulics calculations are understandably needed to obtain a more precise average temperature value for each material, especially in the core. The temperature of multifoil insulation, reflector, B_4C, and MA-956 steel in this study was maintained at 300 K.

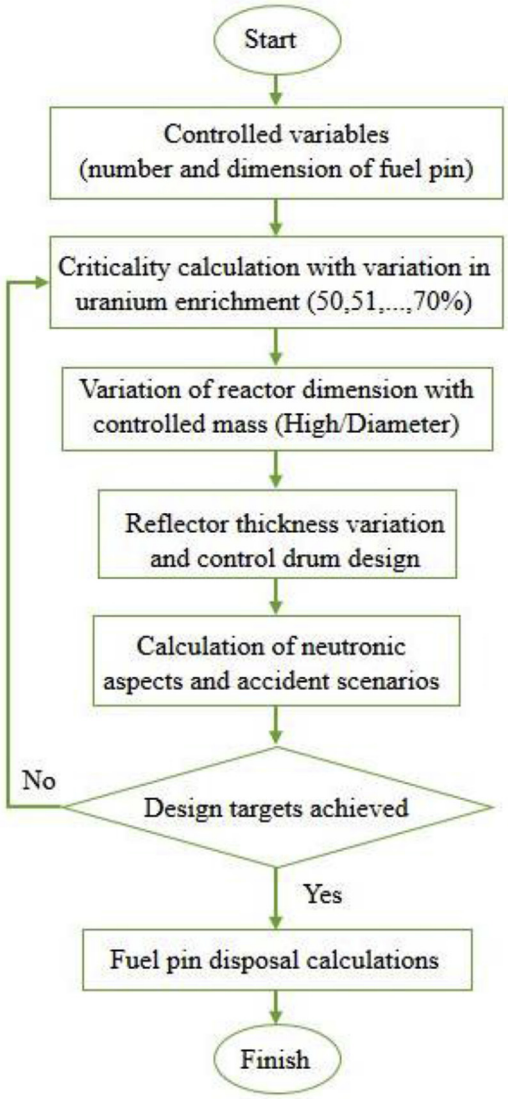


Fig. 5. Reactor design workflow.

In the calculation of fuel temperature coefficient reactivity, the density of the UN would change with temperature according to Eq. (1). In order to keep a constant mass of the UN, the addition of UN volume due to expansion was modeled with high fuel changes, ignoring radial direction expansion. The Na-78 K density would also change depending on the temperature according to Eqs. (2)–(4). The density of other materials was assumed constant as temperature changes, so the density data at 25 °C as presented in Table 1 was still used.

3. Results and discussions

3.1. Criticality at cold clean and operating conditions

The cold clean terminology, in this case, is a fresh fuel condition and the reactor is evaluated at 25 °C. Calculation of reactivity in cold clean conditions is vital to carry out safety analysis during a launch accident. When an accident occurs and the reactor is submerged in water, the reactor remains subcritical. Table 3 presents several ISR criticality calculations in cold clean and operating conditions. As seen in Table 3, reactor submerged in freshwater during an accident has larger reactivity as compared to submerged in seawater.

The difference in the value of the density of H-1 at freshwater and seawater is only 0.4% (i.e., 6.67531×10^{-2} atoms/barn-cm for

Table 3
Reactivity of the reactor during operation and shutdown.

Reactor state	Case	Control drums	UN with natural nitrogen	UN with 100% N-15
			Reactivity (pcm)	Reactivity (pcm)
Operational (temperature 850 K)	Normal	Out	3276 ± 10	6524 ± 10
		In	−5584 ± 10	−2051 ± 10
Cold clean with Mo-30Re SSA	Normal	Out	4023 ± 9	7238 ± 9
		In	−4852 ± 9	−1329 ± 10
	Flooded by seawater	In	−2905 ± 10	371 ± 10
	Flooded by freshwater	In	−2859 ± 9	436 ± 10
Cold clean with Stainless Steel	Flooded by seawater	In	984 ± 10	4271 ± 10
	Flooded by freshwater	In	1175 ± 10	4473 ± 10

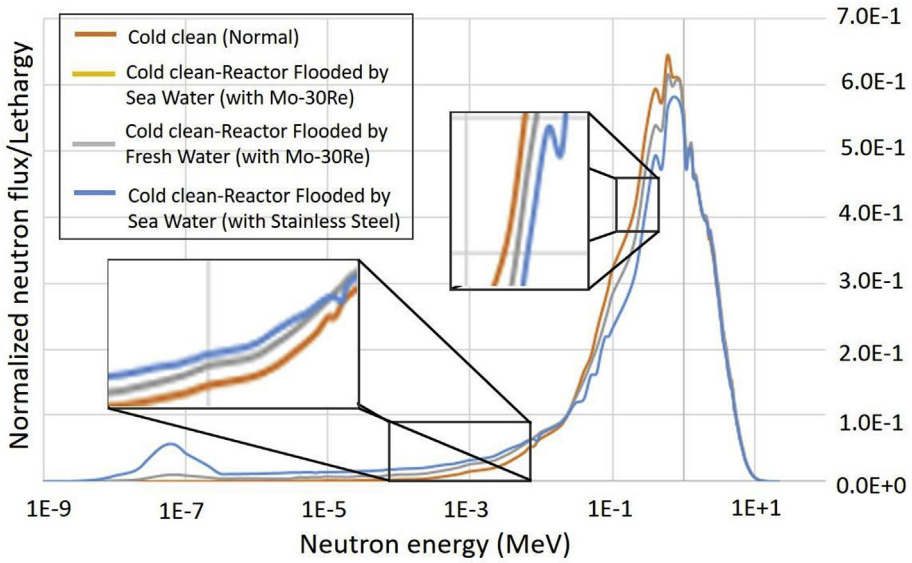


Fig. 6. Neutron spectrum in the ISR core.

freshwater and 6.64901 × 10^{-2} atoms/barn-cm for seawater), and the small amount of water in the reactor (i.e., about 2.6% of the total core volume) makes no significant difference in moderating properties between freshwater and seawater. The difference in reactivity values in these two conditions is only 46 pcm. As seen in Fig. 6, there is no significant differences between the neutron flux in the freshwater-flooded core and the seawater-flooded core (the yellow line mostly coincides the gray line).

As seen in Table 3, reactor submerged in freshwater (H_2O) during an accident has larger reactivity of 46 pcm as compared to submerged in seawater, so that submerged by freshwater is categorized as a more conservative assumption. However, due to the sea surface area is 2/3 of the total surface area of the earth, the likeliness of being submerged in seawater following a launch accident is thus larger than in freshwater. For this reason, we consider the calculation during an accident in both fresh- and seawater (see Section 3.5).

In the design process, ISR was designed with U-235 enrichment of 50, 51, …, 70%, and the value of 55% is selected for the current design. The excess reactivity of ISR at 55% enriched U-235 is 4023 pcm or ~6.267$ (with the delayed neutron fraction of 0.00642). The ideal space reactor should have at least an excess reactivity at the cold clean condition of around 5$. For example, the S4 reactor has reactivity of 4$ at cold clean with enrichment U-235 of 58.8% and it is designed to provide a power of 550 kW_{th} for 7 years of equivalent full power operation (King and El-Genk, 2006). Reducing the enrichment of U-235 to 54% in the ISR leads to an excess reactivity at cold clean of 5.024$, which is roughly ideal. However, it does not reach the reactor design target for the operating time of 10 years and the minimum power of 0.5 MW_{th}. On the other hand, increasing the enrichment to 56% makes the ISR to have shutdown margin of 1550 pcm when the worst accident to occur, and this also does not meet the design target of 2000 pcm. The shutdown margin less than 2000 pcm is feared will cause recriticality accident. The use of 55% enriched U-235 also to reduce the NPT issues and fuel production costs.

Control drums "out" refers to the surface of B_4C facing out of the reactor and "in" refers to facing the reactor. In normal cold clean condition (i.e., no accident), ISR has ρ_{excess} of 4023 ± 9 pcm and shutdown margin of 4852 ± 9 pcm. The accident condition was modeled on two cases: (a) using Mo-30Re alloy as SSA material, and (b) using stainless steel, instead of Mo-30Re alloy. These two cases were intended to test the functionality of Re and Mo-30Re SSAs. It can be seen that the ISR with Mo-30Re remained subcritical when it is flooded by freshwater and seawater. On the other hand, the flooded ISR without Mo-30 Re is supercritical. These results prove the effectiveness of using Mo-30Re alloy as SSA. Under operating conditions, ISR had reactivity of 3276 ± 10 pcm and −5584 ± 10 pcm on the state of the control drums out and in, respectively.

Natural nitrogen contains 99.64% N-14 and 0.36% N-15; hence it may be worthwhile to consider the use of pure N-15 instead of natural nitrogen. When ISR uses UN with 100% N-15, an increase in reactivity occurs due to smaller N-15 absorption cross-section as compared with that of N-14. It is seen that the use of 100% N-15 in ISR is better on the neutron economy. However, it can also be seen in Table 3 that the use of 100% N-15 in ISR leads to positive reactivity when the reactor is submerged in the water. Moreover, increasing N-15 content obviously increases the cost. A thorough fuel cycle cost calculation has not been

performed yet in our current work, so at the current state, it cannot be determined whether the excellence of neutron economy outweighs the fuel cost. For these two reasons, natural nitrogen is selected in this design.

For applications in nuclear reactors, the N-14 causes the presence of C-14 by a neutron-proton reaction ($^{14}_{7}N + n \rightarrow {}^{14}_{6}C + p$). When using the UN with natural nitrogen, the ISR produces C-14 of 7.27×10^{12} Bq in the 10th year and 1.02×10^{13} Bq at the end of reactor lifetime. When using UN with 100% enrichment N-15, the ISR produces C-14 of 3.45×10^{9} Bq in the 10th year and is 1.11×10^{10} Bq at the end of reactor lifetime. However, the presence of more C-14s when using UN with natural nitrogen is not a biologically interesting issue, because ISR is designed as a space reactor for deep space exploration missions and there are no human crew or other living creatures inside the vehicle.

Fig. 6 displays neutron flux in ISR core in four conditions with all control drums out: (a) normal condition (orange line), (b) reactor flooded by seawater and using Mo-30Re (yellow line), (c) reactor flooded by freshwater and using Mo-30Re (gray line) and (d) reactor flooded by seawater and using stainless steel (blue line). In these four conditions, fast neutron still dominates in the core. In condition (b), (c) and (d) there are decreasing in a fraction of fast neutron and increasing in a fraction of epithermal and thermal neutron due to moderation by seawater. The increase in epithermal and thermal neutron fractions under conditions (b) and (c) is lower than conditions (d) because neutron radiative capture cross-section of SSAs Mo-30Re is greater in the thermal neutron spectrum (see Fig. 1), while neutron radiative capture cross-section of stainless steel tends to be relatively flat in the range of thermal energy to fast. The nature of SSAs in the Mo-30Re is what keeps the reactor subcritical. The yellow line coincides with the gray line, indicating no significant difference in the neutron flux distribution between submerged by seawater and freshwater.

3.2. Reactivity coefficients

Reactivity feedback is always an important part of reactor safety. The reason for safety requires that reactivity coefficients be negative. This becomes inherent safety in the reactor. Changes in reactivity values due to changes in temperature at the reactor are caused by changes in the neutron cross-section values due to Doppler broadening and thermal expansion material which results in changes in density.

Fig. 7 shows that the ISR's temperature reactivity coefficient is -0.7872 pcm/K. This value is larger than the temperature reactivity coefficient value of Monju and EBR-II fast reactors of -1.09 to

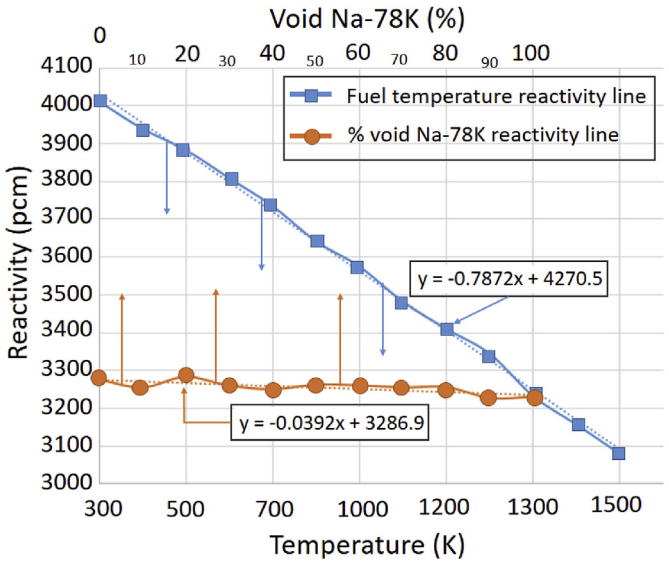


Fig. 7. Temperature and void Na-78 K reactivity coefficients.

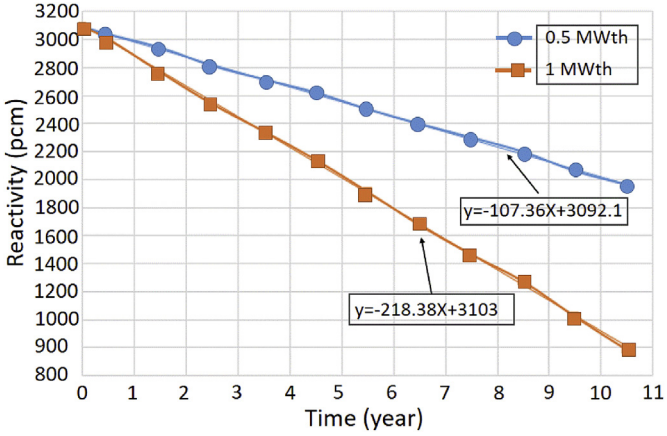


Fig. 8. Reactivity as a function of time.

-1.42×10^{-3} pcm/K and -1.31×10^{-3} pcm/K (Kasahara et al., 2017; Yevick, 1996).

Void may develop in Na-78 K coolant as the reactor operates, either due to an accident or a problem in the cooling system. The value of the ISR's void reactivity coefficient is -0.0392 pcm/%.

3.3. Burnup analysis

The ISR burn-up calculation is performed at the power level of 0.5 MW_{th} and 1 MW_{th}, and all control drums placed in a most critical position. The graph of the relationship between reactivity versus time is shown in Fig. 8. It can be seen that changes in the ISR reactivity versus time are linear at both power levels. Basically, ISR is not designed as a breeder reactor. In addition, this is because ISR uses HEU as fuel in which the weight percentage of U-235 is larger than that of U-238, so the transmutation rate of $^{238}_{92}U \rightarrow {}^{239}_{94}Pu$ can be balanced by the nuclide growth rate which has a high radiative capture value such as Xe-135 and Sm-149.

Changes in the ISR reactivity versus time can be approximated by linear equations of $y = -107.36x + 3092.1$ for the power of 0.5 MW_{th}, and $y = -218.38x + 3103$ for the power of 1 MW_{th}, where the y-axis refers to reactivity in pcm and the x-axis for time in year. At the end of year 10, the ISR still has a reactivity of 2018.5 pcm and 919.2 pcm at the power level of 0.5 MW_{th} and 1 MW_{th}, respectively. By extrapolating the linear equations, it is estimated that the ISR's reactivity will be close to zero in the 28th and 14th years, for power level 0.5 MW_{th} and 1 MW_{th}, respectively, implying it meets the design target.

3.4. Control drum worth

Fig. 9 shows the workflow of the control drum design. Dozens of variation have been made to get the design that achieves the design target. B_4C thickness of 1.5 cm with 80% enriched boron-10 makes the ISR remains subcritical when the worst accident occurs with shutdown margin of 2859 pcm (see Table 3). Shutdown margin value is 1001 pcm if the thickness of B_4C of 1 cm and boron-10 enrichment of 90% are used. This margin does not meet the design target and might cause a re-criticality accident.

The orientation of control drum relative to the core is denoted as θ°. The value of $\theta^{\circ} = 0$ starts from the condition of the control drum facing the core or control drum in. The control drum is then rotated at the value of θ°. Fig. 10 shows the integral and differential curves of reactivity worth in one control drum, out of total 6 drums. The results show the value of the ISR's integral one control drum worth of -1420 pcm. Because the geometry of the ISR is symmetrical in a radial direction, (see Fig. 4), so the total value of the ISR integral control drum worth is equal to 6×-1420 pcm $= -8520$ pcm. Changes in the value

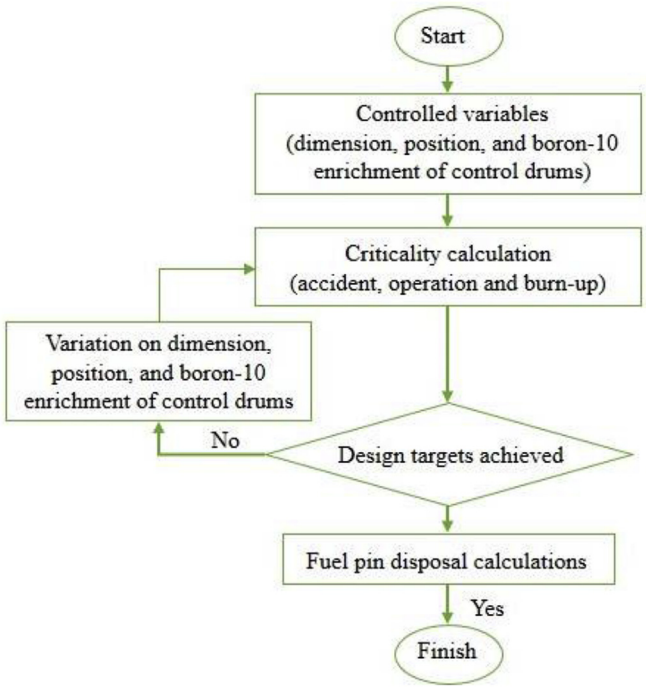


Fig. 9. Control drums design workflow.

of the largest reactivity occur at angle $\theta^{\circ} = 90$ and 300 of ± 414 pcm (differential one control drum worth) or ± 2484 pcm for 6 control drums.

It can be seen that the failure of 1 control drum that has an integral worth of −1420 pcm will have fatal consequences. Actually, the amount of control drum in ISR can be added to minimize the worth 1 control drum, keeping the total worth constant. Consequently, adding the number of control drums will increase the number of movable components in the reactor. Given the space conditions that have a resultant gravity force close to zero, a little force can change the direction of movement of the spacecraft. It can be seen that the addition of the number of control drums will be good neutronically, but it may increase the potential for interference spacecraft movement during space travel. The use of a lesser amount of movable components is preferable in a spacecraft. For example, KRUSTY developed by NASA only uses 1 rod as a safety rod (Mencarini and King, 2018).

The explanation above shows a paradox in the design of space reactor. So the use of 6 control drums on ISR is considered to be the best design. With total 1 control drum failure, ISR will stop operating entirely. It is important to note that the ISR should be kept subcritical when this situation occurs. Further discussion on this matter will be explained in section 3.5.

3.5. Launch failure accident scenarios

Several accident scenarios of ISR in cold clean conditions have been calculated. As previously considered in Section 3.1, all cases describe that the reactor is flooded by either freshwater or seawater, for example in case of launch failure and the reactor falls into water body such as the sea, river and lake. Fig. 11 shows the ISR radial cross-section during accident scenarios and the calculation results are presented in Table 4.

When there is a total failure of 1 control drum on ISR, it is not expected that the reactor will continue to operate. So the main thing to do is to keep the reactor stable and subcritical. From the case (a), it can be concluded that the failure of 1 control drum is still relatively safe (see Table 4). Failure of 2 control drums makes the reactor remain subcritical both in case (b) and case (c) except for the case (b) flooded by freshwater. Case (b) presents a very small shutdown margin that it will be vulnerable to re-criticality conditions. The conclusion that can be drawn from cases (a) - (c) is that the maximum number of failed control drums during the reactor flooded accident is 1.

When more than 1 control drum is damaged, it can be anticipated by applying the cases (d) and (e) schemes. The release or removal of one reflector segment can compensate for up to 2 other damage control drums, both in flooded by freshwater and seawater. In case (f), the reactor reactivity value is negative but very small margin on case flooded by seawater and also vulnerable to re-criticality, but positive when flooded by freshwater. It can be concluded that the release of one reflector segment can only compensate for up to 2 damage control drums. Reflector segment which is discarded is certainly the part that has a damaged control drum, and of course, it will be a disadvantage if the removed one is a reflector segment that has a good control drum, instead. For the scenario of discharged reflector segment, neutron leakage to the environment must be of particular concern. On the other hand, an emergency situation like this is certainly a better choice than leaving a supercritical reactor until core melts or explodes.

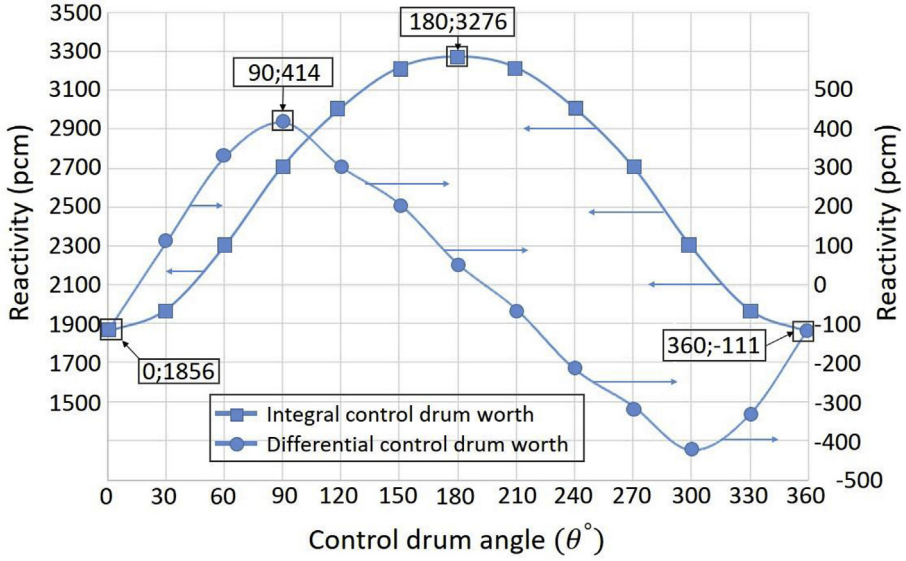


Fig. 10. Reactivity worth of one control drum.

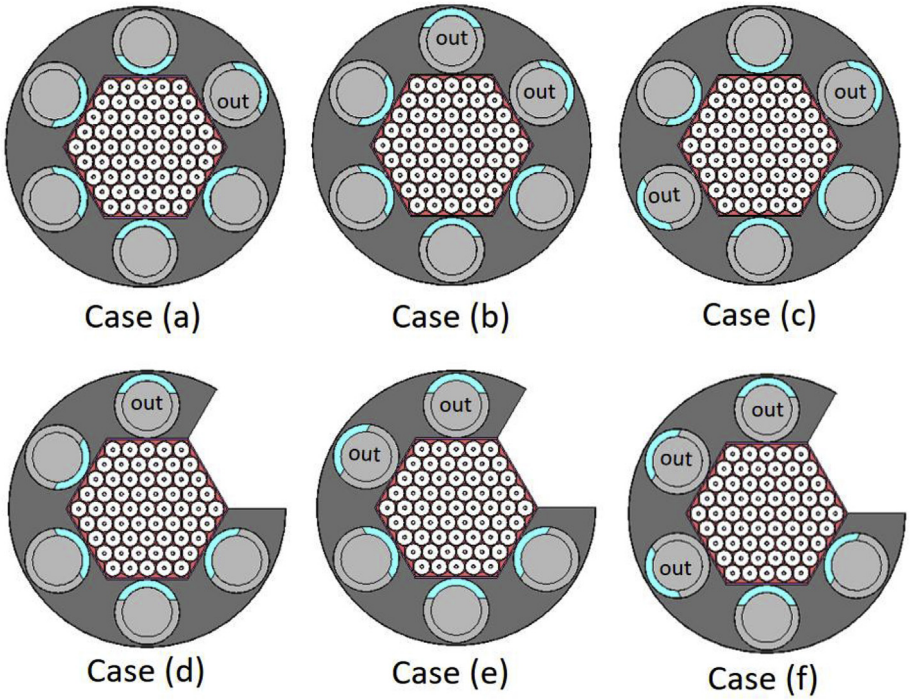


Fig. 11. ISR radial cross-section in several accident scenarios of cold clean and flooded by either freshwater or seawater.

Table 4
Neutronic performance of the ISR in several accident scenarios of cold clean and flooded by seawater and freshwater.

Case	Control drums	Reflectors segment	Reactivity (pcm)	
			Flooded by Seawater	Flooded by Freshwater
(a)	1 broken	Intact	−1524 ± 10	−1493 ± 10
(b)	2 broken (number 1 and 2)	Intact	−15 ± 9	18 ± 10
(c)	2 broken (number 1 and 4)	Intact	−287 ± 10	−241 ± 10
(d)	1 broken	1 dismantled	−3256 ± 10	−3151 ± 10
(e)	2 broken	1 dismantled	−1825 ± 10	−1727 ± 10
(f)	3 broken	1 dismantled	−68 ± 10	12 ± 10

3.6. Operational accident scenario

Simulation of accident scenarios is carried out at the reactor operating condition. This scenario illustrates that when the reactor operates, there are severe failure or damage to the cooling channels that may lead to melting of the fuel pin or core due to the failure of the heat transfer process. In order to overcome such an event, the damaged fuel pin should be discarded or removed from the reactor system. When a damaged fuel pin is removed, the reactor will resume operating normally because each of the other fuel pins has an individual cooling channel that is still operating normally.

Every scenario in this section are done at the beginning of life (BOL) and all control drums out. Table 5 shows the results of neutron calculations for this accident scenario. It can be seen that if 1 to 2 fuel pins are removed, the reactor remains critical. It is also apparent that the disposal of the fuel pins will reduce the reactor operational life. Using a simple calculation from the results of previous calculations in Section 3.3, for power 0.5 MW_{th} and 1 MW_{th} annually requires reactivity about 107 pcm and 218 pcm. If those values are used, it is estimated that the reactor age with 1 discarded fuel pin is 17 years and 8.4 years for 0.5 MW_{th} and 1 MW_{th}, respectively, whereas for 2 discarded fuel pins, the age is 4.4 years and 2.2 years.

Table 5
Neutronic calculation of ISR operational accident scenarios.

Reactor	Reflectors segment	Control drums	Fuel pin	Reactivity (pcm)
Operating	Intact	Out (all)	1 dismantled	1832 ± 10
			2 dismantled	471 ± 10
			3 dismantled	−931 ± 9

3.7. ISR future design

After the calculation of the neutronic conceptual design of ISR is done and described in this article, then detailed thermal hydraulics calculations need to be performed. The results of thermal hydraulic calculations can be used as a basis for designing the dimension of each inner cooling channels. The purpose of doing so is to flatten the heat distribution in a radial direction.

The use of tungsten as SSAs is also a promising option, considering that economically the price of tungsten is cheaper than rhenium. Because the tungsten density and the neutron absorption cross-section of tungsten are lower than rhenium, the use of tungsten as SSAs on ISR might change the ISR dimension significantly.

The use of liquid Na-78 K at room temperature requires pumps in the cooling system and consequently any movements of the pumps will certainly give noise to the spacecraft. Another promising alternative to the cooling system is sodium heat pipe. The use of heat pipes can eliminate the requirements for pumps as it is based on a passive cooling

system. In addition, the use of sodium can increase the reactor operating temperature, because its boiling point is higher than Na-78 K. The large number of pins in the ISR will certainly result in the same amount of Na heat pipe. A large number of heat pipes will make the reactor system more complex. The solid Na phase at room temperature makes the use of Na heat pipes require an additional pre-heating system.

In general, the operational scheme such as described in Section 3.6 will produce at least two issues: (a) it is required to have robust mechanical, electronic and control systems for fuel pin disposal, (b) a scheme like this will cause additional radioactive waste in space. As with the reflector segment removal scenario, such extreme steps are forced to be taken to avoid severe accidents.

4. Conclusions

Neutronic calculations for a conceptual design of Indonesian Space Reactor (ISR) have been carried out. ISR uses Na-78 K coolant with fast neutron spectrum and is designed with a minimum power 0.5 MW_{th} and operating time at full power $\geq$10 years. The ISR hexagonal core is composed of 61 fuel pins. The ISR fuel pin is designed in the form of a hollow cylinder so that each fuel pin has an individual cooling channel. This design aims to make the heat distribution in the core more evenly distributed with the scheme of each fuel pin having different cooling mass flow rates. The temperature of the reactor inlet and outlet is 750 K and 950 K. UN with natural nitrogen and enrichment U-235 of 55% is used as fuel and a total mass of 155 kg.

In cold clean conditions, ISR has excess reactivity of 4023 $\pm$ 9 pcm. Under operating conditions, ISR's excess reactivity and shutdown margin are 3276 $\pm$ 10 pcm and $-$5584 $\pm$ 10 pcm. The use of rhenium and Mo-30Re alloys as SSAs in ISR is advantageous in the accident conditions within which the reactor is flooded by water. The reactivity values when the reactor shuts down and is flooded by cold freshwater when using Re and Mo-30Re as SSAs is $-$2859 $\pm$ 9 pcm whereas the reactivity of the reactor without SSAs is 1175 $\pm$ 10 pcm.

The values of ISR temperature reactivity coefficient and void Na-78 K reactivity coefficient are $-$0.7872/K and $-$0.0392/%void, respectively. Each control drum has a reactivity worth of $-$1420 pcm and $-$8520 pcm for all 6 control drums. Changes in the value of the largest reactivity occur at an angle $\theta^{\circ} = 90$ and 300 of $\pm$ 414 pcm for one control drum and $\pm$ 2484 pcm for 6 control drums. At the end of the 10th year, ISR still has excess reactivity of 1958 pcm and 785 pcm at the power level 0.5 MW_{th} and 1 MW_{th}. It is estimated that ISR will be subcritical in the 28th and 14th years at power levels 0.5 MW_{th} and 1 MW_{th}.

In the scenarios of the reactor flooded by cold water (either freshwater or seawater), the maximum number of failures of the control drum is 1. When more than 1 control drums are damaged, then removal of the reflector segment is the best solution to keep the reactor subcritical. When 1 reflector segment is removed, then the maximum number of damaged control drums to keep the reactor still subcritical is 2.

The scenario of removing fuel pins that are damaged during operation is a challenge for further research in terms of mechanics, electronics and control, and the addition of radioactive waste in space. Simulations carried out during operating at the beginning of life demonstrate that a maximum of 2 fuel pins can be disposed to keep the reactor critical.

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